

BharatAssist: Simplifying Government for Every Citizen

Project Proposal for UCS503P

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1.Higher-order goal

This project aims to improve citizen access to public services by reducing the time, effort, and confusion involved in navigating government procedures and understanding legal documents. While the broader objective is to promote transparent, efficient, and citizen-centric governance, the immediate engineering objective is to transform complex bureaucratic processes into an accessible, measurable, and deployable software solution.

2.Time-to-value

- We will deliver a functional prototype within the first two weeks that enables users to search government services, navigate application procedures, and simplify legal documents using AI.
- We will prioritize rapid iterations and continuous user feedback by implementing weekly feature releases, supported by CI-based automated testing and deployment.
- Success in the first iteration will be measured by delivering a usable workflow that allows citizens to quickly find service information and understand legal documents, with subsequent iterations improving accuracy, usability, and feature coverage based on user feedback.

3.Problem Statement

Citizens in India often face significant delays, confusion, and uncertainty while accessing government services and understanding legal documents. Information is scattered across multiple government websites, offices, and departments, making it difficult to identify the correct procedures, required documents, eligibility criteria, and timelines. Additionally, legal documents are frequently written in complex language that is difficult for the average citizen to understand. Common pain points include:

- **Information Fragmentation:** Government procedures, forms, eligibility requirements, and service details are distributed across multiple websites and departments, forcing citizens to spend considerable time searching for accurate information.
- **Complex Legal Language:** Legal notices, agreements, certificates, and government documents are written using technical and legal terminology, making them difficult for non-experts to understand and increasing the risk of errors or misinterpretation.
- **Lack of Personalized Guidance:** Citizens often do not know the appropriate government office to approach, the sequence of steps to follow, or the documents required for a specific service, leading to repeated visits and application delays.
- **Limited Transparency:** Users have limited visibility into application procedures, expected processing times, and common reasons for rejection, resulting in uncertainty and reduced trust in public service delivery.
- **Inefficient Access to Public Services:** Existing government portals primarily provide static information and lack an intelligent, citizen-centric interface that simplifies navigation, answers queries, and guides users throughout the service process.

Why this matters now (contextual relevance): This project addresses the everyday challenges citizens face in accessing government services and understanding complex legal documents. By providing a unified, AI-powered platform, it reduces confusion, saves time, and makes public services more accessible and user-friendly.

4. Proposed Solution

4.1 Overview

- Build a web-based AI-powered Bureaucracy Navigator and Legal Document Simplifier with the following components:
- Service Navigator: Provides step-by-step guidance for government services, including eligibility, required documents, fees, and procedures.
- Government service information will be sourced from official government portals and updated periodically
- Legal Document Simplifier: Converts complex legal and government documents into simple, easy-to-understand language.
- AI Assistant: Answers user queries and recommends relevant government services.
- Admin Dashboard: Manage service information and legal document resources.

4.2 Core Workflow

- User searches for a government service or uploads a legal document.
- The system provides AI-generated guidance or a simplified explanation.
- User receives the required documents, application steps, and relevant information to complete the process.

4.3 Operational Constraints

- Ensure the platform is responsive and works efficiently on low-to-mid range devices.
- Use existing AI models and web technologies instead of developing new algorithms.
- Design the system for easy deployment using cloud hosting and managed databases.

This project focuses on building a scalable, AI-powered platform using existing technologies to simplify government service navigation and legal documents. The emphasis is on usability, workflow automation, rapid iterations, and CI/CD rather than developing new AI algorithms.

5. Solution Approach

5.1 AI Assistance & Quality Assurance (non-research)

- Use existing AI/LLM models to simplify legal documents and answer government service queries.
- Validate user inputs and recommend relevant government services based on keywords.
- Present AI-generated responses as guidance, allowing users to verify information through official government sources.
- AI-generated responses are provided as guidance and should be verified through official government sources.
- Ground all AI responses using a RAG pipeline that retrieves relevant verified data (government service info or uploaded document content) before generation, rather than relying on the LLM's parametric knowledge alone.

5.2 Web Architecture (web-based solution)

A typical 3-tier web architecture:

- Frontend: Responsive interface for users to search government services, upload legal documents, and interact with the AI assistant.
- Backend (Flask): Handles authentication, AI requests, service recommendations, and document processing.
- Database (MySQL): Stores user profiles, government service information, legal document metadata, and user history

5.3 CI/CD and Fast Delivery Enabler

CI/CD will be used to:

- Automatically build and test the application on every code commit.
- Deploy updates quickly through an automated deployment pipeline.
- Enable frequent feature releases and continuous user feedback.

6. Evaluation Criterion

We define success using measurable metrics related to accessibility, usability, and user experience.

6.1 Primary Evaluation Metric

- Information Retrieval Time (IRT): Average time taken by a user to find the required government service information or receive a simplified legal document.
- Target: Average IRT ≤ 2 minutes during the pilot phase.
- Attribution: Measured using application logs (search request to result display)

6.2 Secondary Evaluation Metrics

- Document Comprehension Score: Average user rating ($\geq 4/5$) on how easy the simplified document is to understand.
- Search Success Rate: Percentage of users who successfully find the required government service information on their first attempt.
- User Satisfaction: Average user feedback rating of $\geq 4/5$ for usability and guidance.
- System Reliability: Application availability of $\geq 99\%$ during the testing period.

6.3 Pilot Validation Plan (high-level)

- Recruit 10–20 pilot users (students/citizens) to test government service navigation and legal document simplification.
- Measure information retrieval time, user satisfaction, and document comprehension before and after using the platform through surveys and application logs.

7. Scalability

The proposed solution is designed to scale both technically and operationally.

1. Scalable (theoretically):

The system will support increasing users and government services by:

- separating the frontend, backend, and AI modules,
- using efficient database indexing and caching,
- adopting a stateless API architecture.

2. Operationally deployable (practically)

The system can be deployed using common cloud platforms with:

- a managed database,
- containerized or cloud-based deployment,
- a CI/CD pipeline for automated testing and deployment.

8. Engine availability heuristic

The project will leverage mature technologies and services for implementation, including:

- HTML, CSS, Bootstrap, and JavaScript for building a responsive and user-friendly web interface.
- Flask (Python) for backend services, REST APIs, and AI integration.
- MySQL for storing user accounts, government service information, and legal document metadata.
- Self-hosted open-source LLM (Llama 3.1 8B, served locally via Ollama) for AI-powered legal document simplification and government service guidance, ensuring citizen data and legal documents remain within our own infrastructure rather than being sent to third-party APIs.
- Retrieval-Augmented Generation (RAG) pipeline using ChromaDB (vector database) and sentence-transformers (all-MiniLM-L6-v2) for embeddings, to ground AI responses in verified government service data and the actual uploaded document — reducing hallucination and improving factual accuracy.
- Session/JWT-based authentication for secure user access.
- GitHub Actions for CI/CD, automated testing, and deployment.

We will use these mature technologies to focus on workflow design, usability, and rapid iterative development rather than developing new algorithms or infrastructure.

9. Project scope and deliverables

9.1 Initial deliverables

- Government service search with step-by-step guidance.
- AI-powered legal document simplification.
- User authentication and profile management.
- Responsive web interface with basic search history.
- CI pipeline for automated build and testing.
- Initial deployment to a cloud hosting platform.

9.2 Subsequent deliverables

- AI chatbot
- RTI drafting
- Multilingual support
- Government scheme recommendations
- Document checklist generation
- Application tracking

10.Risks and Mitigations

- **Data Privacy:** Use a self-hosted, private LLM (no third-party API calls) so citizen data and legal documents never leave our own infrastructure; store only essential user information and securely handle uploaded legal documents.
- **AI Inaccuracy:** Mitigate hallucination using a RAG pipeline that grounds responses in verified government service data and the source document; additionally display AI-generated responses as guidance and encourage users to verify information from official government sources.
- **Changing Government Policies:** Regularly update service information from official government portals.
- **Deployment & Testing:** Use CI/CD pipelines to automate testing and ensure reliable deployments.

11.Summary

This project proposes an AI-powered web platform that helps citizens navigate government services and understand complex legal documents. By combining AI assistance with a centralized knowledge base, the platform improves accessibility, reduces information search time, and supports efficient public service delivery through rapid iterations and CI/CD practices.